

PrefAgent Report

System Overview

PrefAgent is a Python command line application that performs preference based reasoning over sets of binary-attribute objects. It supports penalty logic and qualitative choice logic, and provides five reasoning tasks: Encoding, Feasibility Checking, Show the Table, Exemplification, Omni-optimization.

Architecture

Organized into four modules:

[formula.py](#)- tokenizes, parses, and evaluates propositional formulas

[parser.py](#)- reads all four input files and returns structured python objects ready for the engine

[engine.py](#)- generates all 2^n objects, checks feasibility against hard constraints, computes penalties, computes QCL ranks, and finds optimal objects under each logic

[main.py](#)- interactive program that prompts the user to select a logic type and a task, then it will dispatch to the engine and print results

Logic Details

Penalty Logic: Each rule is a formula paired with a numeric valued penalty. An object receives a penalty when the formula is false. The objects are ranked by total penalty with less penalty being better. This then produces a total order so every object is comparable.

Qualitative Choice Logic: Each rule has the form- $\phi_1 \text{ BT } \phi_2 \text{ BT } \dots \text{ BT } \phi_n \text{ IF } \psi$. An object's rank under a rule is 0 if it does not satisfy the condition ψ , k if it satisfies ψ and ϕ_k is the first option it matches, or $n+1$ if it satisfies ψ but none of the options. Comparison is aggregated across all rules: object A is strictly preferred over B if it is not worse on any rule and strictly better on at least one.

Testing Results

Encoding: All 256 objects were generated and labeled o0(zero)-o255 using the bit position encoding. Object o0 maps to and o255 maps to.

Feasibility: 36 feasible objects were identified. Objects violating any hard constraint are excluded.

Penalty Logic - Table and Optimisation: The penalty table was computed for all 36 feasible objects. The two omni-optimal objects are o191 and o255 both with a total penalty of 7.

Exemplification: o115 vs o74 both scored a total penalty of 20 - equivalent.

Qualitative Choice Logic - Table and Optimisation: The QCL rank table was computed for all 36 feasible objects. The four omni-optimal objects are o66, o74, o200, o202 - none of which is dominated by any other feasible object. Exemplification: o185 vs o119 - o119 is preferred on the action-context and theater-context rules, while o185 is preferred on the family-context rule. Neither dominates, so the result is: incomparable.

Screenshots below showcasing all program features.

```
(base) matthewbosworth@Matthews-Laptop-4 CAP-Project3 % python3.11 main.py
Welcome to PrefAgent!
Enter Attributes File Name: ExampleTestCase/attributes.txt
Enter Hard Constraints File Name: ExampleTestCase/constraints.txt
```

```
Choose the preference logic to use:
1. Penalty Logic
2. Qualitative Choice Logic
3. Exit
Your Choice: 1
You picked Penalty Logic
Enter Preferences File Name: ExampleTestCase/penaltylogic.txt
```

```
Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 1
o0 - ice-cream, beer, beef
o1 - ice-cream, beer, fish
o2 - ice-cream, wine, beef
o3 - ice-cream, wine, fish
o4 - cake, beer, beef
o5 - cake, beer, fish
o6 - cake, wine, beef
o7 - cake, wine, fish
```

```
Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 2
Yes, there are 6 feasible objects.
```

```
Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 3
```

encoding	fish AND wine	wine OR cake	total penalty
o0	10	6	16
o1	10	6	16
o4	10	0	10
o5	10	0	10
o6	10	0	10
o7	0	0	0

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 4

Two randomly selected feasible objects are o0 and o6,
and o6 is strictly preferred over o0.

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 5

All optimal objects: o7

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 6

Choose the preference logic to use:

1. Penalty Logic
2. Qualitative Choice Logic
3. Exit

Your Choice: 2

You picked Qualitative Choice Logic

Enter Preferences File Name: ExampleTestCase/qualitativechoicelogic.txt

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 1

o0 - ice-cream, beer, beef
o1 - ice-cream, beer, fish
o2 - ice-cream, wine, beef
o3 - ice-cream, wine, fish
o4 - cake, beer, beef
o5 - cake, beer, fish
o6 - cake, wine, beef
o7 - cake, wine, fish

Choose the reasoning task to perform:

1. Encoding

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 2

Yes, there are 6 feasible objects.

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 3

	encoding	fish BT beef IF	ice-cream BT cake IF	wine BT beer IF fish	beer BT wine IF beef
o0	2	1	0	1	
o1	1	1	2	0	
o4	2	2	0	1	
o5	1	2	2	0	
o6	2	2	0	2	
o7	1	2	1	0	

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 4

Two randomly selected feasible objects are o4 and o7,
and o4 and o7 are incomparable.

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 5

All optimal objects: o0, o1, o7

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 6

Choose the preference logic to use:

1. Penalty Logic
2. Qualitative Choice Logic
3. Exit

Your Choice: dm

Wrong Choice! Enter your choice: 3

Bye!

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